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The Persistence of Sex Bias in High-Impact Clinical Research

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Abstract

Background: Sex bias is present in clinical research resulting in disparities in the treatment of women. Our objective was to identify the prevalence of sex inclusiveness of participants in human clinical trials after the passage of NIH and U.S. Congress policies in 2015 and 2016 to increase female enrollment in clinical research.

Methods: We performed an observational analysis of data from registered clinical trials published in 3 high-impact biomedical journals from January 1, 2015 to December 31, 2019.

Results: 1,442 manuscripts with 4,765,783 human subjects were included for analysis. Significantly more males (56%) than females (44%) were included in all 3 journals (P<0.0001). Sex matching ≥80% was found in 24.6% of publications. Industry funded 43.7% of all studies enrolling significantly more males than females (60.8% vs. 39.2%, P<0.0001). NIH funded 10.2% of studies enrolling significantly more females than males (52.7% vs. 47.3%, P<0.0001). North America and Europe contributed 82.6% of the studies with each enrolling significantly more males than females (P<0.0001). The United States was the country contributing the most studies (36.1%), enrolling significantly more males than females (55.5% vs. 45.5%, P<0.0001). Cardiovascular disease was the subject area of the most manuscripts among medical specialties (19%), enrolling significantly more males than females (64.9% vs. 35.1%, P<0.0001). Studies analyzed by clinical trial phase, type, trial, and allocation enrolled significantly more males than females (P<0.0001).

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None.

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Conclusions: Sex bias remains prevalent in human clinical research trials. Improvements have been made in NIH-funded clinical trials; however, this constitutes a small percentage of overall studies.

Keywords

Sex Bias; Clinical Research; Disparity; Women

INTRODUCTION

Sex bias has been prevalent in biomedical research for decades and has negatively impacted disease detection, treatment, and outcomes in women.[1,2] In 1994, The Institute of Medicine Committee referenced two types of sex bias that have been present in scientific and observational research.[2] The first type was *Male Bias*, which is when scientists and observers adopt male perspectives, and the second was *Male Norm*, which is when male characterization is accepted as the standard. In 2014 the Food and Drug Administration (FDA) also investigated reasons for the persistent lack of female representation in biomedical research and found causes including investigator and sponsor perceptions that recruitment of female participants required more time and money, and that family responsibilities could limit female participants' abilities to commit for the duration of the study.[3] These biases and flawed perceptions have resulted in unnecessary harm to female patients. A historical example of sex bias is *The Physician's Health Study* in 1982, which enrolled 22,071 male physicians and no female physicians.[4–6] An example of harm that can result from unequal sex representation is in the U.S. General Accounting Office report in 2001, which found that 8 of 10 FDA-approved drugs were withdrawn from the market between 1997-2000 due to greater health risk to women compared with men.[7] There are many examples in which females are poorly represented as research participants across a variety of medical specialties and there is a clear necessity to increase female inclusion in biomedical research.[8–13]

Several initiatives have been developed to create equal representation of male and female participants in biomedical research studies.[14–17] In 1985, The Public Health Service Task Force on Women's Health Issues made recommendations to increase female enrollment in biomedical and behavioral research and to focus on the effect of diseases over the course of a woman's life.[14] This resulted in the development of the Office of Research on Women's Health (ORWH) in 1990 and passage of the Revitalization Act in 1993, which aimed to improve female and underrepresented minorities' inclusion in clinical trials funded by the National Institutes of Health (NIH).[15,17,18] In 2015, the NIH created a policy that made it mandatory for NIH-funded studies to recognize sex as a biological variable and to take sex into account in the design and statistical analysis of the results.[16,19] In 2016, the 21st Century Cures Act mandated the NIH make information publicly available regarding the study population of NIH-funded clinical research, update guidelines on the inclusion of women and minorities in clinical research, and develop guidelines on reporting analysis by sex/gender, race, and ethnicity in Phase III clinical trials.[20–22]

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The efforts to address the ongoing sex bias in biomedical research have resulted in some improvements. Females now represent 50% or more of the total research participants in NIH-funded clinical studies.[16] However, our prior examinations have found that sex bias still persists in biomedical research.[23–26] Specifically, these studies found that in surgical research involving cell, animal, and human participants sex bias favors male sex in representation, data reporting and statistical analysis. Our most recent study compared improvement in surgical research published 2011-2012 to that published 2017-2018, and found that despite the mandated policies, sex bias persists and has improved very little.[25] In this study, our objective was to determine if sex bias is prevalent in human clinical research trials published in high-impact journals 2015-2019. We aimed to identify the prevalence of sex enrollment in human clinical trials relating to total enrollment and to determine the degree of sex matching of participants in each study. We hypothesize that despite the enactments made to increase female enrollment in biomedical research, sex bias is still prevalent overall in human clinical trials as current mandates are focused solely on NIH-funded research.

METHODS

Data Abstraction.

Data were abstracted as previously described by Yoon et al.[23] All original manuscripts involving human clinical research that allowed for the enrollment of eligible male and female participants published in the *Journal of the American Medical Association (JAMA), The Lancet*, and *The New England Journal of Medicine* between January 1st, 2015 and December 31st, 2019 were reviewed by 3 abstractors. Manuscripts that underwent secondary analyses were included for data abstraction, as secondary analyses may not have the same patient enrollment from the initial study. Manuscripts that did not involve human participants, did not have a registered clinical trial identifier, did not specify sex, and those that studied sex-specific diseases and conditions (testicular cancer, prostate cancer, cervical cancer, uterine cancer, and pregnancy) were excluded (Supplemental Figure 1).

Variables Abstracted.

The following data were abstracted from each article: publication date, publication journal, clinical trial identifier, funding, sponsor name, specialty, conditions studied, sex-specific condition, study phase, study allocation, study type, published manuscript title, study dates, sex eligibility for study, location, number of participants and type of trial. For studies that were conducted in multiple continents and countries, the continent and country of the corresponding author's address was used as the primary location.

Sex Matching.

Sex matching was calculated for included manuscripts as previously described by Mansukhani et al.[24] Briefly, a study that is 100% sex-matched and enrolled 150 participants would have enrolled 75 participants of each sex (75/75 X 100 = 100%) and a 50% sex-matched study with 150 participants would have 50 participants of one sex and 100 participants of the other sex (50/100 x 100 = 50%).

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Statistical Analysis.

Pairwise comparisons were obtained from a one-sample chi-squared test that was FDR-adjusted for multiple comparisons or different factors which may affect the proportions of male versus female participants, including sex matching, journal, publication year, funding, continent, country, medical specialty, phase, allocation, type, and trial. Odds ratios were calculated comparing the number of male participants from manuscripts published in 2015 to the number of male participants from manuscripts published in 2019. Results were obtained from a multivariate logistic model with journal, year, and their interaction as predictors and sex as the response. Statistical significance was defined by a P<0.05. Analyses were conducted in SAS version 9.4 (SAS Institute Inc.).

RESULTS

Study Population.

A total of 1,707 manuscripts were abstracted. 265 manuscripts were excluded: 14 manuscripts did not include a clinical registration identifier, 20 studies did not specify sex, and 231 were sex-specific studies. This resulted in 1,442 manuscripts with 4,765,783 enrolled human participants for data analysis. The total number of male participants was 2,646,081 (56%) and the total number of female participants was 2,119,702 (44%, P<0.0001).

Sex Matching.

The most common sex-matching range was 60-69% and was found in 14.4% (208/1442) of manuscripts with 73% (151/208) of these studies containing more male than female participants (Table 1, Supplemental Figure 2). Sex matching at 80% or greater was found in only 24.6% (355/1442) of manuscripts. We also found that 69.6% (1000/1436) contained more male participants with 99% to <5% sex matching (Table 1).

Sex by Journal and Publication Year.

The total enrollment of each abstracted journal had significantly more male than female participants (P<0.0001) (Figure 1). In addition, the total overall enrollment among all 3 journals over the course of the 5 abstracted years showed significantly more male than female participants (P<0.0001). Among all 5 abstracted years from all 3 journals, only *The Lancet* in 2019 had more female participants. Among all 3 journals, *The New England Journal of Medicine* contributed the most manuscripts at 45% (649/1442).

Each journal was examined to see if there was an improvement over the course of the 5 abstracted years relating to the prevalence of males using 2015 as the reference. An odds ratio >1 indicated a higher male prevalence and an odds ratio <1 indicated a higher female prevalence. *JAMA* had a significantly higher prevalence of male participants in 2019 (OR 1.152; 95% CI 1.136-1.168; P<0.0001). *The Lancet* and *The New England Journal of Medicine* had a higher prevalence of female participants in 2019 (OR 0.67; 95% CI 0.664-0.676; P<0.0001) and (OR 0.773; 95% CI 0.767-0.779; P<0.0001), respectively.

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Sex by Funding Source.

Funding for studies was grouped into 5 categories: industry, NIH, United States Federal Agencies, other (agencies from countries other than the United States and charitable organizations), and none (Figure 2). Industry funded the most studies at 43.7% (630/1442) and enrolled significantly more males 1,064,462 (60.8%) than females 687,063 (39.2%; P<0.0001). The NIH funded 10.2% (147/1442) of the studies and enrolled significantly more females 476,723 (52.7%) than males 428,582 (47.3%; P<0.0001). Male predominance was also found in the studies funded by multiple agencies (15.2%; 219/1442) (Supplemental Figure 3).

Sex by Continent and Country.

Reviewed manuscripts were published in a total of 6 continents. The majority of the manuscripts published (1191/1442, 82.6%) were from two continents, North America and Europe. Of the 6 continents, 5 continents published manuscripts that included significantly more male participants (P<0.0001**;** Supplemental Figure 4). Only Africa had studies with significantly more female participants (P<0.0001), but represented only 2.5% (37/1442) of the total manuscripts. Publication location was further analyzed by examining manuscripts published in individual countries (Table 2). A total of 59 countries published manuscripts and of these 59 countries, the United States and United Kingdom contributed the most manuscripts, 36.1% (521/1442) and 12.9% (186/1442) respectively, with each country enrolling significantly more male participants (P<0.0001). Of the remaining 57 countries, no country contributed more than 6.3% (91/1442) of the total manuscripts. 13.5% (8/59) of countries published manuscripts that had no significant difference in the number of male and female participants (P>0.05) and 20.3% (12/59) had significantly more female participants (P<0.05); however, these countries accounted for 3.9% (56/1442) of the total manuscripts.

Sex by Medical Specialty.

Manuscripts were published by 55 different medical specialties (Table 3; Supplemental Figure 5). Of these 55 specialties, only 10.9% (6/55) had manuscripts that did not have a significant difference in the number of male and female participants (P>0.05). 29.1% (16/55) had a majority of female participants, with 13 being significant (P<0.05). Cardiovascular disease was represented in the most manuscripts at 19% (274/1442) with significantly more males 1,074,001 (65%) than females 581,453 (35%; P<0.0001).

Sex by Study Phase, Allocation, Type, and Trial.

Study phase was broken down into Phase 1, Phase 2, Phase 2/3, Phase 3, Phase 4, Phase N/A and Phase Not Listed (Supplemental Figure 6). Phase 3 encompassed the most studies at 42.9% (620/1442). All phases had significantly more male participants (P<0.0001). Study allocation was broken down into randomized, non-randomized, N/A, and randomized/non-randomized. Randomized trials made up the majority of the study allocation at 92.9% (1340/1442) with significantly more male participants (P<0.0001; Supplemental Figure 7). Study type was broken down into interventional, observational, and patient registries (Supplemental Figure 8). Interventional studies made up the majority of studies at 98.4% (1420/1442) and enrolled significantly more male participants (P<0.0001).

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Study trial was broken down into pharmacological, device, and other, with some studies having a combination (Supplemental Figure 9). 68.7% (992/1442) of the study trials were pharmacological and enrolled significantly more male participants (P<0.0001).

DISCUSSION

Our results show that sex bias in human clinical trials published in high-impact journals remains prevalent despite legislation and policies to improve female participation. Sex matching to at least 80% was found in less than a quarter of the studies. Each of the 3 abstracted journals published studies with significantly more male than female participants. Industry funded the most studies and had a significantly higher number of male participants, while NIH-funded studies had significantly more female participants. North America contributed the most studies by continent and the United States by country, with each having significantly more male participants. Studies analyzed by phase, allocation, type, and trial all enrolled significantly more male participants. Efforts made to increase female participants in NIH-funded studies have had a positive effect, however, studies funded by other agencies continue to lag behind in female participants. Despite the evidence that women and men experience diseases and respond to therapies differently, and the requirement for NIH-funded studies to consider sex as a biological variable in study design, results and analysis, there has not been overall improvement in female representation in clinical trials aside from NIH-funded research.

When examining the population of the continents and countries that contributed the most studies, each enrolled significantly more male subjects despite an almost equal distribution of sex in the general population. Only the United States and Canada contributed studies from North America, with females making up 50.8% and 50.3% of the total populations respectively.[27,28] Canada has had several initiatives to improve female in enrollment in biomedical research beginning with the 1997 policy, Inclusion of Women in Clinical Trials During Drug Development.[29] In December 2010, the Canada Institute of Health and Research initiated a policy for those applying for grants to specify if investigators are considering gender and/or sex and to provide justification.[30] In 2013, the Canadian government updated its policies on the inclusion of women in clinical trials and on providing an analysis of sex differences.[31] Europe and the United Kingdom each produced male majority studies, but also have near equal sex distribution in the general populations of 51% and 50.6% respectively.[32] In 2014, the European Union created guidelines on creating gender balance and the integration of gender and sex analysis in biomedical research for those applying for funding.[33] The United Kingdom National Institute for Health Research began an investigation in 2017 into the underrepresentation of individuals in clinical trials called the Innovations in Clinical Trial Design and Delivery for the Under-served (INCLUDE).[34,35] The INCLUDE project provides guidance for researchers, reviewers, and funding agencies on the inclusion of underrepresented populations in clinical trials.

Cardiovascular disease was overwhelmingly under represented by female participants at 35% of all research participants. Yet, cardiovascular disease is the number one cause of death in women and cardiovascular disease affects almost an equal number of males and females from the age of 60-79 and affects more women than men from the age of 89-90.[36,37]

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Interestingly, the surgical specialty that had the largest sex discrepancy was Cardiothoracic Surgery, with women making up 37% of research participants.

It is evident that medications, medical devices, and various diseases affect men and women differently. Due to this knowledge there have been multiple attempts to improve the sex bias in biomedical research; however, there remains room for improvement. In 2015, Geller et al. compared their findings to their previous 2004 study examining randomized control studies funded by the NIH in the reporting of sex and ethnicity. The median enrollment of women was 43% in 2004 and improved to 46% in 2015; however, only 26% of studies analyzed in 2015 included sex in their analysis.[38,39] Following this study in 2016, the NIH required applicants to consider sex as a biological variable in research design, analysis and reporting.[19] Subsequently, the ORWH released data on the enrollment for NIH-funded studies in 2017 and 2018 and saw an improvement in female enrollment from 47.2% to 52.4% respecively.[40] We found similar female enrollment in NIH studies with female participants making up 52.7% of enrolled participants from 2015 to 2019. Although female enrollment in NIH-funded clinical research studies was shown to be equitable to males, we found that NIH-funded studies comprised only 10.2% of the clinical trials studies published in 3 high-impact journals from 2015 to 2019. Industry-funded clinical research comprised 43.7% of studies and female participants made up only 39.2% of research participants. Thus, there is a great need to develop policies that pertain to non-NIH funded research. Regardless of funding source, all new medications must be approved by the FDA through the New Drug Application, prior to marketing. It is during this process that a medication is evaluated for its efficacy, safety and pharmacological profile. Thus, it is through new regulations by the FDA that widespread change can result. FDA regulations that mandate the recognition of sex as a variable and the importance of designing clinical trials and analyzing and discussing results based on sex can have a profound impact on all human clinical research, irrespective of funding source.

Our research group previously investigated sex bias in all interventional clinical trials published on clinicaltrials.gov from 2013 to 2015.[41] When comparing the clinicaltrials.gov data to the manuscripts published in 3 high-impact journals, a significant difference in sex enrollment favoring males based on study phase and study allocation remains. This is surprising given that studies published in high-impact journals likely undergo much more scrutiny and may be more likely to balance enrollment of both sexes better. Sex matching did improve slightly among the clinical trials published in the high-impact journals, with >50% and >80% sex-matched studies increasing from 56.6% to 63.9% and 22.2% to 24.6% respectively, compared to the clinicaltrials.gov data. Interestingly, industry-funded clinical research studies made up the majority of studies from the clinicaltrials.gov analysis at 83.7% and males made up 50.7% of participants. In our current study that is limited to those published in 3 high-impact journals, industry funded only 43.7% of the clinical research studies but there was a larger discrepancy between male (61%) and female (39%) enrollment. The regulations and mandates to improve female representation appear to have no bearing on private industry, and as such, there remains a large sex disparity in this area. To make a more significant impact, the changes that were instituted by the United States Congress and the NIH also need to encompass industry-funded clinical research.

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The *2019-2023 Trans-NIH Strategic Plan for Women's Health Research* addressed the ongoing issues that have resulted in the paucity of medical research regarding women's health, ranging from women in the United States having a lower life expectancy compared to other high-income countries, to the rising maternal mortality rate.[42] One of the goals of the strategic plan was to educate researchers on the effect sex and gender has in biomedical research. As a result, the ORWH created an online course that is designed to help biomedical researchers create and analyze studies to address sex as a variable.[43] In addition, the Sex and Gender Equity in Research (SAGER) Guidelines were created as a way to standardize sex and gender reporting in biomedical research.[44]

On another front, the International Committee of Medical Journal Editors (ICJME) made recommendations to include analysis based on sex as a variable when reporting results.[45] Of the 3 journals from which we abstracted data, *JAMA* and *The Lancet* ask authors to account for sex as a variable.[46,47] *The New England Journal of Medicine* released an editorial discussing the lack of diversity in clinical trials and beginning January 1st, 2022 manuscripts submitted are required to have a supplemental table providing information on disease, problem or condition and the representativeness of the study group.[48] We previously examined sex-based reporting in surgical research and found that first and senior female authors are more likely than their male counterparts to include female participants in their studies.[26] It is important for all those involved in conducting and publishing biomedical research to be aware of the ongoing sex bias in biomedical research and the impact that sex bias has had on the diagnosis, treatment and outcomes in women.

Our study does contain limitations. We examined the enrollment of male and female participants; however, we did not analyze the results and discussion to identify if authors conducted an analysis based on sex or addressed differences in sex representation. We abstracted data regarding the final participants' numbers enrolled in the clinical trials, but we did not abstract the overall initial recruitment data before exclusion criteria were applied. It is possible more women were excluded from studies based on inclusion/exclusion criteria or declined enrollment. Reform on female research participant enrollment began in 1985, however, the aim of our study was to look at the changes after the mandates made in 2015 in 2016. Studies that were designed and enrollment that was completed prior to these mandates may not accurately reflect the recent directives. We did not identify other possible variables that could have changed male and female enrollment over time. Lastly, we could not differentiate between gender and sex based on the reported data.

CONCLUSION

Despite recognizing the importance of sex inclusiveness in clinical research and the recommendations and requirements that have been enacted, sex bias remains prevalent in biomedical research. Improvements have been made in NIH-funded clinical trials, however, this makes up a small percentage of overall studies published. With the inclusiveness seen in NIH-funded studies after regulations were enacted, there is a clear need to address industry-funded clinical research studies and create similar mandates to increase female representation. Eliminating sex bias in biomedical research will also require the entire

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biomedical research community, including authors, peer-reviewers, journal editors, and funding agencies, to hold each other accountable for sex as a biological variable.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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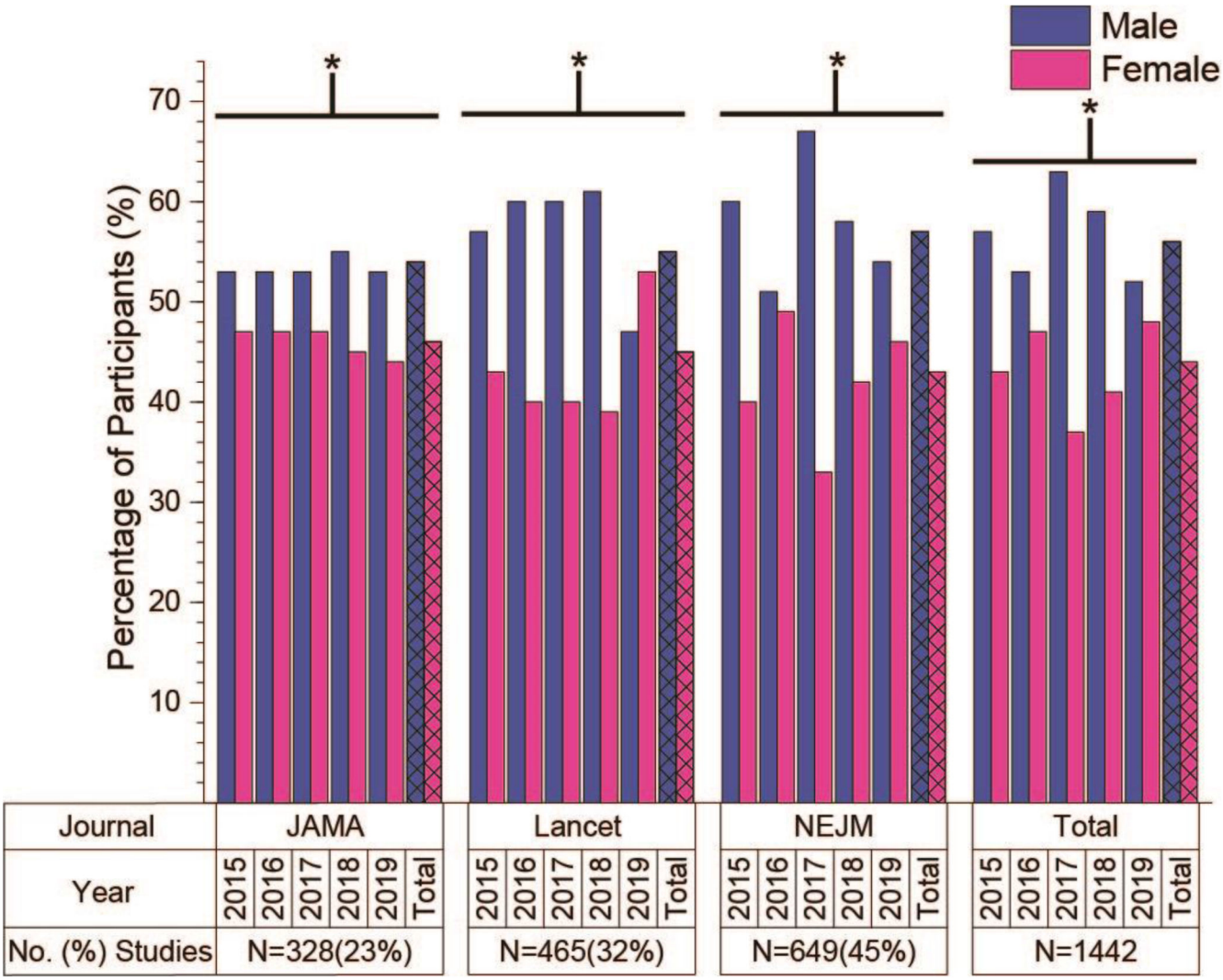

Figure 1.

Sex Enrollment by Journal and Publication Year.

Comparison of *JAMA*, *The Lancet* and *The New England Journal of Medicine* enrollment of participants by individual year and total enrollment for the respective journal from 2015 to 2019 followed by total enrollment of all journals for each individual year and total enrollment for all journals from 2015 to 2019. *P<0.0001

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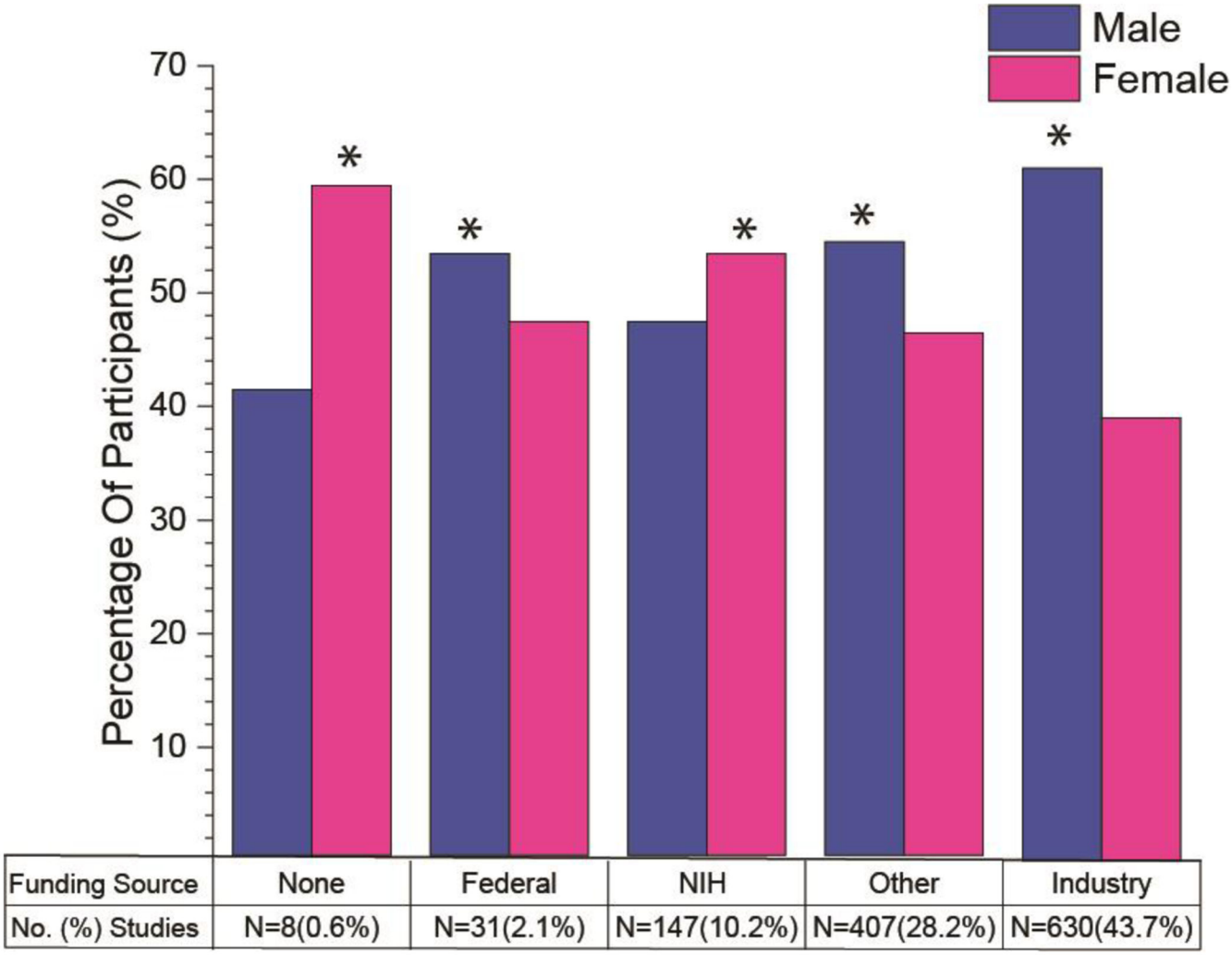

Figure 2.

Sex Enrollment Based on Study Funding by a Single Source.

Comparison of sex enrollment by funding sources. *P<0.0001

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Table 1.

Sex Matching of Study Participants in Clinical Trials Published in 3 High-Impact Journals 2015-2019

Degree of Sex Matching	Total Studies No. (%)*	Male Majority Studies No. (%)	Female Majority Studies No. (%)
99-90%	158 (11)	93 (59)	65 (41)
89-80%	191 (13)	119 (62)	72 (38)
79-70%	171 (12)	114 67)	57 (33)
69-60%	208 (14)	151 (73)	57 (27)
59-50%	188 (13)	131 (70)	57 (30)
49-40%	134 (9)	104 (78)	30 (22)
39-30%	170 (12)	137 (81)	33 (19)
29-20%	124 (9)	89 (72)	35 (28)
19-10%	63 (4)	46 (73)	17 (27)
9-5%	13 (1)	6 (46)	7 (54)
<5%	16 (1)	10 (63)	6 (38)

*The total number of studies equals 1436. 6 Studies had 100% sex matching

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Table 2.

Studies Contributed and Sex Enrollment by Country 2015-2019*

Country	Total Studies No. (%)	Male No. (%)	Female No. (%)	P-value
United States	521 (36.1)	1,182,735 (55)	950,181 (45)	<0.0001
United Kingdom	186 (12.9)	301,001 (57)	226,671 (43)	<0.0001
France	91 (6.3)	48,891 (59)	33,718 (41)	<0.0001
Canada	89 (6.2)	171,232 (60)	113,736 (40)	<0.0001
Germany	89 (6.2)	100,291 (59)	68,798 (41)	<0.0001
Australia	77(5.3)	96,537 (58)	70,072 (42)	<0.0001
Netherlands	67 (4.6)	33,821 (60)	22,808 (40)	<0.0001
Japan	39 (2.7)	36,957 (64)	21,098 (36)	<0.0001
China	25 (1.7)	32,389 (53)	28,568 (47)	<0.0001
Spain	23 (1.6)	26,279 (60)	17,458 (40)	<0.0001
Switzerland	21 (1.5)	18,267 (65)	9,798 (35)	<0.0001
Italy	18 (1.2)	27,951 (73)	10,345 (27)	<0.0001
Denmark	17 (1.2)	60,734 (67)	30,216 (33)	<0.0001
Belgium	17 (1.2)	8,231 (59)	5,690 (41)	<0.0001
Sweden	13 (0.9)	14,981 (63)	8,729 (37)	<0.0001
India	12 (0.8)	37,593 (51)	36,401 (49)	0.00002
South Korea	9 (0.6)	7,025 (69)	3,204 (31)	<0.0001
Brazil	9 (0.6)	21,684 (46)	25,923 (54)	<0.0001
South Africa	9 (0.6)	15,914 (33)	32,870 (67)	<0.0001
Argentina	8 (0.6)	40,215 (72)	15,271 (28)	<0.0001
Norway	8 (0.6)	7,924 (68)	3,739 (32)	<0.0001
Finland	7 (0.5)	2,591 (54)	2,242 (46)	<0.0001
Russia	7 (0.5)	11,571 (51)	11,334 (49)	0.13
New Zealand	6 (0.4)	2,829 (58)	2,014 (42)	<0.0001
Taiwan	5 (0.3)	1,429 (54)	1,209 (46)	0.00002
Uganda	5 (0.3)	74,598 (45)	92,084 (55)	<0.0001
Israel	4 (0.3)	2,836 (74)	988 (26)	<0.0001
Poland	4 (0.3)	5,627 (64)	3,188 (36)	<0.0001
Hungary	4 (0.3)	4,912 (51)	4,733 (49)	0.08
Ukraine	3 (0.2)	567 (64)	325 (36)	<0.0001
Greece	3 (0.2)	580 (63)	343 (37)	<0.0001
Congo	3 (0.2)	1,481 (58)	1,058 (42)	<0.0001
Nigeria	3 (0.2)	2,778 (53)	2509 (47)	0.0003
Burkina Faso	3 (0.2)	19,733 (51)	18,591 (49)	<0.0001
Indonesia	3 (0.2)	5,782 (51)	5,533 (49)	0.02
Vietnam	3 (0.2)	39,281 (46)	45,798 (54)	<0.0001

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Country	Total Studies No. (%)	Male No. (%)	Female No. (%)	P-value
Austria	3 (0.2)	784 (32)	1,656 (68)	<0.0001
Thailand	2 (0.1)	210 (71)	84 (29)	<0.0001
Saudi Arabia	2 (0.1)	1,722 (59)	1,175 (41)	<0.0001
Kenya	2 (0.1)	2,270 (56)	1,750 (44)	<0.0001
Sierra Leone	2 (0.1)	300 (52)	272 (48)	0.25
Bangladesh	2 (0.1)	95,006 (46)	110,770 (54)	<0.0001
Bulgaria	2 (0.1)	195 (43)	263 (57)	0.002
Qatar	1 (0.07)	1,362 (83)	282 (17)	<0.0001
Singapore	1 (0.07)	279 (76)	90 (24)	<0.0001
Portugal	1 (0.07)	118 (69)	54 (31)	<0.0001
Zambia	1 (0.07)	117 (56)	92 (44)	0.09
Tanzania	1 (0.07)	16,783 (52)	15,216 (48)	<0.0001
Nepal	1 (0.07)	10,264 (51)	9,755 (49)	0.0004
Malawi	1 (0.07)	36,277 (51)	34,514 (49)	<0.0001
Niger	1 (0.07)	2,406 (50.1)	2,392 (49.9)	0.84
Iran	1 (0.07)	3,398 (49.7)	3,440 (50.3)	0.62
Guinea	1 (0.07)	244 (49)	258 (51)	0.55
Chile	1 (0.07)	198 (47)	226 (53)	0.19
Liberia	1 (0.07)	2,957 (45)	3,675 (55)	<0.0001
Botswana	1 (0.07)	3,582 (40)	5,392 (60)	<0.0001
Cameroon	1 (0.07)	209 (34)	404 (66)	<0.0001
Ghana	1 (0.07)	75 (27)	204 (73)	<0.0001
Zimbabwe	1 (0.07)	78 (14)	495 (86)	<0.0001
Total	**1442 (100)**	**2,646,081 (56)**	**2,119,702 (44)**	

*Sorted by number of manuscripts published by country, highest to lowest.

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Table 3.

Studies Contributed and Sex Enrollment by Medical Specialty 2015-2019*

Medical Specialty	Total Studies No. (%)	Male No. (%)	Female No. (%)	P-value
Cardiovascular Disease	275 (19.1)	1,074,397 (65)	581,580 (35)	<0.0001
Medical Oncology	184 (12.8)	170,914 (54)	144,411 (56)	<0.0001
Infectious Disease	119 (8.3)	389,477 (47)	435,084 (53)	<0.0001
Neurology	93 (6.4)	61,249 (46)	72,613 (54)	<0.0001
Critical Care Medicine	71 (4.9)	65,600 (58)	46,827 (42)	<0.0001
Pulmonary Disease	63 (4.4)	54,789 (52)	50,574 (48)	<0.0001
Endocrinology, Diabetes & Metabolism	58 (4)	81,008 (61)	52,526 (39)	<0.0001
Pediatrics	44 (3.1)	19,893 (56)	15,554 (44)	<0.0001
Rheumatology	44 (3.1)	17,216 (39)	27,287 (61)	<0.0001
Gastroenterology	39 (2.7)	18,733 (55)	15,169 (45)	<0.0001
Hematology	38 (2.6)	31,743 (53)	28,573 (47)	<0.0001
Internal Medicine	25 (1.7)	15,860 (51)	15,203 (49)	0.0002
Psychiatry	25 (1.7)	58,317 (55)	48,308 (45)	<0.0001
Surgery	25 (1.7)	261,163 (46)	311,666 (54)	<0.0001
Public Health & General Preventive Medicine	23 (1.6)	147,859 (51)	140,292 (49)	<0.0001
Allergy & Immunology	22 (1.5)	5,593 (55)	4,616 (45)	<0.0001
Dermatology	21 (1.5)	9,553 (63)	5,654 (37)	<0.0001
Ophthalmology	21 (1.5)	3,944 (46)	4,562 (54)	<0.0001
Nephrology	19 (1.3)	8,696 (59)	5,941 (41)	<0.0001
Hepatology	17 (1.2)	3,649 (64)	2,096 (36)	<0.0001
Neonatal-Perinatal Medicine	17 (1.2)	10,023 (46)	11,545 (54)	<0.0001
Cardiothoracic Surgery	16 (1.1)	26,487 (67)	12,816 (33)	<0.0001
Orthopedic Surgery	16 (1.1)	5,605 (49)	5,929 (51)	0.003
Pediatric Critical Care Medicine	14 (1)	5,427 (53)	4,771 (47)	<0.0001
Vascular Surgery	13 (0.9)	17,247 (71)	7,065 (29)	<0.0001
Emergency Medicine	12 (0.8)	11,389 (53)	10,172 (47)	<0.0001
Anesthesiology	10 (0.7)	5,204 (51)	5,029 (49)	0.09
Colon & Rectal Surgery	9 (0.6)	2,512 (59)	1,753 (41)	<0.0001
General Practice	9 (0.6)	4,094 (46)	4,757 (54)	<0.0001
Physical Medicine & Rehabilitation	8 (0.6)	9,129 (59)	6,333 (41)	<0.0001
Transplant Surgery	8 (0.6)	1,296 (65)	685 (35)	<0.0001
Geriatric Medicine	7 (0.5)	8,794 (55)	7,126 (45)	<0.0001
Neurological Surgery	7 (0.5)	2,610 (48)	2,833 (52)	0.003
Pain Medicine	7 (0.5)	1,656 (59)	1,133 (41)	<0.0001
Pediatric Hematology-Oncology	7 (0.5)	667 (48)	729 (52)	0.10
Addiction Medicine	6 (0.4)	2,366 (61)	1,508 (39)	<0.0001

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Medical Specialty	Total Studies No. (%)	Male No. (%)	Female No. (%)	P-value
Otolaryngology	6 (0.4)	940 (65)	510 (35)	<0.0001
Radiation Oncology	6 (0.4)	1,440 (63)	844 (37)	<0.0001
Surgical Oncology	6 (0.4)	2,880 (56)	2,242 (44)	<0.0001
Urology	6 (0.4)	2,172 (77)	632 (23)	<0.0001
Pediatric Emergency Medicine	4(0.3)	1,477 (55)	1,220 (45)	<0.0001
Trauma Surgery	3 (0.2)	1,900 (73)	716 (27)	<0.0001
Adolescent Medicine	2 (0.1)	9,958 (51)	9,633 (49)	0.02
Pediatric Endocrinology	2 (0.1)	1,162 (53)	1,049 (47)	0.02
Pediatric Infectious Diseases	2 (0.1)	2094 (50)	2094 (50)	0.99
Pediatric Rheumatology	2 (0.1)	70 (24)	227 (76)	<0.0001
Sleep Medicine	2 (0.1)	196 (77)	59 (23)	<0.0001
Surgical Critical Care	2 (0.1)	602 (78)	169 (22)	<0.0001
Child & Adolescent Psychiatry	1 (0.06)	124 (85)	22 (15)	<0.0001
Clinical Genetics	1 (0.06)	30 (45)	36 (55)	0.48
Clinical Pharmacology	1 (0.06)	167 (34)	322 (66)	<0.0001
Family Medicine	1 (0.06)	806 (43)	1,076 (57)	<0.0001
Neuroradiology	1 (0.06)	5,605 (49)	5,804 (51)	0.07
Occupational Medicine	1 (0.06)	83 (43)	115 (57)	0.03
Pediatric Gastroenterology	1 (0.06)	216 (50.5)	212 (49.5)	0.86
Total	**1442 (100)**	**2,646,081 (56)**	**2,119,702 (44)**	

*Sorted by number of manuscripts published by medical specialty, highest to lowest.